The Demobilizing Effect of Selective Exposure and Heterogeneous Responses to Violence*

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Abstract

The burden of violence is not shared equally. Yet, our understanding of its consequences often comes from populations unlikely to be victimized. Using principal stratification, I show that because exposure to violence is selective and reactions to it are heterogeneous, effects among the exposed can differ from the population average. Empirically, I reassess the mobilization-spurring consequences of crime. Recovering all causal effects consistent with victimization and participation rates across 94 surveys in 20 Latin American countries, I show population-wide average effects favor mobilization. Conversely, effects among victims are overwhelmingly negative. Panel evidence from Mexico confirms that those likely to demobilize are more exposed to crime. Similar reversals arise when reassessing civil war abduction in Uganda and police contact in the United States. Where selective exposure and heterogeneous reactions coincide, the effect among the exposed is the causal quantity of interest. Targeting this effect reveals that violence likely demobilizes those exposed.

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Empirical social science has converged on the view that treatment effects that average across all units should serve as the benchmark for causal knowledge and policy evaluation. For treatments that could, in principle, reach anyone—vaccination campaigns, cash transfers, educational curriculum changes—population-wide causal contrasts are maximally informative: they reflect what we should expect, on average, at scale. But when it comes to exposure to violence, one of the treatments most studied in political science —criminal victimization, civil war violence, police contact, incarceration, and forced displacement— only specific subpopulations that systematically differ from the rest experience it. Further, we have no normative interest in exposing everyone in the population to them. In such cases, the population average answers a question about a universal exposure counterfactual that would never occur and that no one would wish for. I argue that for treatments where exposure is selective and responses are heterogeneous, and where the treatment is not one we would ever seek to universalize, the relevant causal contrast is instead the effect among those exposed, not the population-wide average.

The participatory consequences of violence are a case where the distinction between these two causal contrasts is consequential, because violence exhibits both selective exposure and heterogeneous responses. Participatory responses to violence range from mobilization through grievance and collective action (Blattman, 2009; Bateson, 2012) to demobilization through fear and withdrawal (Ley, 2018; Skogan, 1986). Further, exposure to different types of violence is routinely selective: criminals target vulnerabilities (Browning, Pinchak and Calder, 2021), police discriminate (Knox, Lowe and Mummolo, 2020), and civil war violence follows territorial logic (Kalyvas et al., 2006). Moreover, the characteristics that predict political participation, such as risk tolerance, mobility, and community embeddedness (Kam, 2012; Gerber et al., 2011), overlap substantially with those that predict victimization.

To characterize the consequences of the coincidence of selective exposure and heterogeneous responses to violence, I develop a principal stratification framework with binary treatment and binary outcome (Frangakis and Rubin, 2002). In contrast to the standard compliance applica-

tion, I define strata by individuals' potential outcomes on the outcome of interest: whether they would participate regardless of treatment, only if treated, only if untreated, or never. This captures response heterogeneity. I then extend the framework to allow the probability of exposure to vary across these response types, capturing overlapping selective exposure. I show that the population-wide average treatment effect (ATE) depends only on the relative shares of those who would respond positively and negatively to treatment, while the average effect among the treated (ATT) also depends on how exposure is distributed across these groups.

Because the ATT weights response types by their exposure probabilities whereas the ATE does not, the two estimands can point in opposite directions. Whether sign reversal occurs, and how large the divergence is, depends on two quantities: the intensity of selection—the degree to which one response type is more likely to be exposed than the other—and the degree of response heterogeneity—the relative prevalence of those who would respond positively versus negatively across the entire population. When exposure is sufficiently skewed toward those who would respond in the opposite direction of the population average, a positive population-average effect masks a negative effect among the exposed. From these results, I derive a practical and generalizable tool for any setting with binary treatment and binary outcome : given an estimated population-average effect and a belief about how differentially selected the exposed are, one can determine whether the effect among the exposed could be of opposite sign. I provide this mapping in Table 2.

I use the framework to reassess recent work on the participatory consequences of criminal violence in Latin America that targets population-wide causal effects. Several influential studies report a positive population-average effect of victimization on political participation (Bateson, 2012; Dorff, 2017; Boulding, Mullenax and Schauer, 2022). This finding has been read as a silver lining: victimization may foster more engaged citizens and, through them, more accountable governance. But this conclusion rests on effects that average over the entire population, most of whom will never be victimized. I investigate whether the direction of this effect is robust to focusing instead on those actually exposed to criminal violence.

To assess whether the mobilization finding survives the shift from the population average to the victimized, I develop a partial identification procedure that bounds both the ATE and the ATT using four cross-sectional moments observable in survey data. I extract these moments from repeated cross-sectional surveys of 20 Latin American countries (LAPOP, 2022), an extended version of the same data from Bateson (2012). The procedure maps every combination of response-type shares and exposure probabilities consistent with the data to a specific pair of effects, revealing the range of plausible values for each and the degree of selective exposure required to generate what we observe. I call each such combination a *parameter vector*: a complete assignment of stratum shares (π_θ) and stratum-specific treatment probabilities (ρ_θ) that reproduces the observed data.

Across four measures of non-electoral participation, 63%–85% of parameter vectors yield a positive ATE, consistent with the population-average effects reported in influential studies. But 53%–93% of those same vectors yield a negative ATT. Under a flat prior over the parameter space, this can be interpreted as the probability that the effect among the victimized is negative. For most of the parameter space consistent with what we observe, victimization demobilizes those who experience it. Extending the analysis to electoral participation shows that different forms of political behavior can be affected by victimization in opposite directions, reflecting differences in the underlying structure of selection and response across outcomes. Specifically, self-reported willingness to vote, the pattern reverses: the ATE is predominantly negative while the ATT is positive.

The cross-sectional evidence establishes that sign reversal is likely. But what is actually happening empirically? I turn to a quarterly rotating panel survey of Mexican city residents to answer this question (INEGI, 2024). Between-individual comparisons show that soon-to-be victims are more mobile, witness more criminal and gang activity, and report more violent conflicts than their neighbors before being victimized. This is evidence of selective exposure: those who will be victimized are systematically different from those who will not. Within-individual comparisons show that after victimization, individuals become less mobile, less trusting of law enforcement, and more

fearful, consistent with demobilization.

Taken together, the panel evidence tells us that demobilizers are more likely to be selected into victimization. The framework tells us that the sign of the ATT depends on whether mobilizers are sufficiently more numerous in the population to overcompensate for their lower exposure. Since the population-average effects in the literature are substantively small, on the order of 1 to 4 percentage points (Bateson, 2012), mobilizers cannot sufficiently outnumber demobilizers to produce a positive ATT, and the effect among the victimized is signed negative for non-electoral participation.

I extend the partial identification analysis to findings on the participatory consequences of civil war abduction in Uganda (Blattman, 2009) and police contact in the United States (Walker, 2020). In both settings, the identified sets exhibit the same structure: population-average effects favor mobilization while effects among the exposed favor demobilization.

The paper makes three contributions. First, I contribute to the literature on the political consequences of violence. Influential studies find that criminal victimization (Bateson, 2012; Dorff, 2017; Boulding, Mullenax and Schauer, 2022), civil war exposure (Blattman, 2009; Bellows and Miguel, 2009), and police contact (Walker, 2020) increase political participation, though the evidence across settings is mixed. I show that across all three settings, the mobilization finding may reflect population-average effects that mask demobilization among those actually exposed. The reversal follows from a structural feature common to all three: the people most likely to experience violence are often also the people most likely to withdraw from political life after experiencing it.

Second, I contribute to work on the epistemology of causal empirical research by emphasizing the interpretative pitfalls of population-wide effects when heterogeneous exposure and response coincide. In the econometrics of program evaluation, Heckman and Vytlacil (2005) showed that no single treatment effect parameter summarizes an intervention when selection into treatment covaries with how individuals respond to it, and Samii, Paler and Daly (2017) proposed estimands tailored to the populations most relevant for policy. Building on this insight, my framework reduces

the question of sign reversal to two interpretable quantities—the intensity of selection (λ) and the degree of response heterogeneity (η)—on which researchers can bring substantive priors, without requiring a fully specified structural model. Any researcher with a binary treatment, a binary outcome, and a belief about how selective exposure is can assess whether conclusions drawn from population averages hold up under the shift to the exposed.

Third, I contribute to work on the accumulation of causal evidence by showing that population-specific estimands incorporate heterogeneity differently, and that this matters for whether findings across studies are commensurable. Researchers have emphasized that empirical estimates must be linked to theoretical claims through an explicit estimand (Lundberg, Johnson and Stewart, 2021) and that for findings to accumulate across studies, estimates must target the same causal quantity (Slough and Tyson, 2023). I show that under selective exposure the estimand simultaneously defines the link between theory and evidence and the population over which effects are averaged (Mesquita and Tyson, 2020). Research designs targeting different estimands are not commensurable even when they examine the same treatment, outcome, and data. The finding that violence mobilizes and the finding that violence demobilizes can both be correct; they answer different questions, defined over different populations.

1 A Principal Strata Framework

In this section, I develop a principal stratification framework that accounts for both heterogeneous responses and selective treatment exposure. The framework makes precise the conditions under which the population-average effect and the effect among the exposed diverge, and yields a partial identification procedure for recovering both from observable data.

Consider a population of units indexed by i and a binary treatment $Z \in 0, 1$, for example, experiencing violence or not. Denote the potential outcome for unit i under treatment assignment z as $Y_i(z)$. Following Frangakis and Rubin (2002), I classify units into principal strata (θ) defined by their joint potential outcomes $(Y_i(1), Y_i(0))$, where the outcome itself serves as the post-treatment

variable with respect to which strata are defined. Thus, by definition, the individual treatment effect for each i, $ITE_i = Y_i(1) - Y_i(0)$, will be identical for all units in the same stratum.

Denote the proportion of units in each stratum π_θ, with $\sum_\theta \pi_\theta = 1$. I depart from the canonical principal-strata setup by allowing units to have stratum-specific probabilities of receiving treatment, ρ_θ. Table 1 synthesizes the setup for the case of a binary outcome $Y_i(z) \in 0, 1$, such as whether an individual participates in a given political activity or not.

Stratum (θ)	**Share**	**Pr(Z=1)**	**Y(1)**	**Y(0)**	**ITE**
Always Participate	π_A	ρ_A	1	1	0
Participate if **T**reated	π_T	ρ_T	1	0	1
Participate if **U**ntreated	π_U	ρ_U	0	1	-1
Never Participate	π_N	ρ_N	0	0	0

Table 1: Stratum-specific individual treatment effects and treatment probabilities. Each row defines a principal stratum by its potential outcomes under treatment and control. The treatment probability ρ_θ is allowed to vary across strata.

For a binary treatment such as exposure to violence, we can classify individuals into four strata: those who always participate (A), those who never participate (N), those who participate if and only if treated (T), and those who participate if and only if untreated (U).[1]

Proposition 1. *The average treatment effect is* $ATE = \pi_T - \pi_U$.

Proposition 1 applies the principal stratification framework of Frangakis and Rubin (2002) to binary outcomes. It states that the ATE depends exclusively on the relative shares of strata T and U. This result is straightforward when we consider that all units in strata A and N contribute an ITE of zero. Consequently, the magnitude and direction of the ATE will only depend on the

[1] I make the stable unit treatment value assumption (SUTVA) throughout.

frequency with which T appears in the population, relative to U.[2]

Proposition 2. *The average treatment effect on the treated is*

$$ATT = \frac{\pi_T \rho_T - \pi_U \rho_U}{Pr(Z = 1)},$$

where $Pr(Z = 1) = \sum_\theta \pi_\theta \rho_\theta$.

Conversely, the ATT, or the average expected change in outcomes for those who are treated, will be a function of population shares and also of the stratum-specific probabilities of receiving treatment. These stratum-specific probabilities scale the relative contributions of the T and U strata within the treated population. Intuitively, strata with higher treatment probabilities constitute a larger share of those observed in the treated group, and therefore should contribute more to the ATT. Further rearranging the numerator of the expression for the ATT yields an identity for the population-level impact of treatment:

$$ATT \times Pr(Z = 1) = \pi_T \rho_T - \pi_U \rho_U. \tag{1}$$

I denote this quantity ΔY, the average change in Y attributable to the realized treatment assignment.

Corollary 1. *The ATE and ATT are equal when* $\rho_T = \rho_U = Pr(Z = 1)$.

Intuitively, when $\rho_T = \rho_U = Pr(Z = 1)$, all strata face the same probability of treatment, so conditioning on being treated does not alter the relative representation of any stratum. That is, the treated group is a random draw from the population, so the ATE and the ATT average over, in expectation, identical weights.[3]

[2]Formal proofs for all expressions in this section are provided in Appendix Appendix A.

[3]Corollary 1 also parallels the condition identified by Heckman and Vytlacil (2005) under which all treatment parameters coincide. They derived this result within a structural framework that models the selection process explicitly,

Corollary 2. *When $\rho_T = \rho_U$, the ATE and ATT have the same sign. Specifically, $ATT = \frac{\rho_T}{Pr(Z=1)} \cdot ATE$.*

Corollary 2 relaxes Corollary 1 by allowing the treatment-insensitive strata (A and N) to have treatment probabilities that differ from those of the treatment-sensitive strata. Because the A and N strata have a treatment effect of zero, their over- or under-representation in the treated group cannot change the sign of the ATT relative to the ATE. Only the relative weighting of T and U determines the sign, so $\rho_T = \rho_U$ suffices for sign agreement. The treatment-insensitive strata do, however, affect the *magnitude*: the scaling factor $\rho_T/Pr(Z = 1)$ can amplify or attenuate the ATT but cannot reverse its direction. Corollary 1 is the special case where $\rho_T = Pr(Z = 1)$, so the scaling factor equals one.

When do the ATE and ATT disagree in sign? The ATE and ATT need not agree in sign. Signs agree when composition and selection reinforce each other: for example, when the T-stratum is both larger than the U-stratum and at least as likely to be treated, or when the U-stratum is both larger and at least as likely to be treated. Signs can diverge when one stratum is larger but the other is more likely to be treated, so that the stratum that dominates the population average is under-represented among the treated. I now characterize exactly when this occurs.

To do so, I define two summary parameters. Let $\lambda = \rho_U/\rho_T$ capture differential selection between the responsive strata. When $\lambda > 1$, the U-stratum is more likely to be treated than the T-stratum; when $\lambda < 1$, the reverse holds. Last, when $\lambda = 1$ we have equal selection and necessarily sign agreement (Corollary 2). Let $\eta = \pi_T/\pi_U$ capture the relative size of the two responsive strata. Hence $\eta > 1$ when the T-stratum is larger ($ATE > 0$) and $\eta < 1$ when the U-stratum is larger ($ATE < 0$).

Proposition 3. *Let $\pi_T > 0$ and $\pi_U > 0$. Then $\text{sign}(ATT) = \text{sign}(\pi_T\rho_T - \pi_U\rho_U)$, and the ATE*

requires an instrument, and recovers the full marginal treatment effect curve. Here, the same condition emerges from the structure of principal stratification alone, without parametric assumptions on the selection process or an exclusion restriction.

and ATT have opposite signs if and only if $\lambda > \eta$ when $ATE > 0$, or $\lambda < \eta$ when $ATE < 0$.

The sign of the ATT is determined by whether the T-stratum's exposure-weighted size ($\pi_T \rho_T$) exceeds that of the U-stratum ($\pi_U \rho_U$). When $\lambda > 1$, the U-stratum is over-represented among the treated, pulling the ATT toward negative values. A sign reversal occurs when this differential selection more than compensates for the size advantage of the T-stratum, that is, when $\lambda > \eta$. The condition is symmetric: when $ATE < 0$ ($\eta < 1$), the ATT is positive only if $\lambda < \eta$. In that case, the T-stratum must be sufficiently more exposed to treatment than the U-stratum to overcome its smaller population share.

Corollary 3. *When $ATE > 0$ and $\lambda > 1$:*

$$ATT < 0 \quad \Longleftrightarrow \quad \pi_U > \frac{ATE}{\lambda - 1}. \tag{2}$$

By symmetry, when $ATE < 0$ and $\lambda < 1$, the ATT is positive if and only if $\pi_T > |ATE| \cdot \lambda/(1-\lambda)$.[4]

For the sign of the ATT to oppose the sign of the ATE, the countervailing stratum must constitute a sufficiently large share of the total population. How large depends on two quantities: (i) the magnitude of the ATE, and (ii) how much more exposed the countervailing stratum is to treatment. When the ATE is small or λ is far from one, the threshold is low, so sign reversal requires only a modest population share of the countervailing stratum. Table 2 illustrates this relationship for a range of ATE magnitudes and selection ratios λ. The table applies to any binary treatment and binary outcome; it follows entirely from the framework and requires no empirical input.

Consider the bottom-left region of the table, where the ATE is positive and the T-stratum is also more likely to be treated ($\lambda < 1$). These cells show check marks: composition and selection reinforce each other, so the ATE and ATT always agree in sign regardless of stratum shares. Now consider the bottom-right region, where the ATE is still positive but the U-stratum is more likely to

[4] Define $\lambda' = \rho_T/\rho_U = 1/\lambda$ and $\eta' = \pi_U/\pi_T = 1/\eta$. The sign-flip condition $\lambda < \eta$ is equivalent to $\lambda' > \eta'$, and the same algebra yields $\pi_T > |ATE|/(\lambda' - 1) = |ATE| \cdot \lambda/(1 - \lambda)$.

be treated ($\lambda > 1$). Here, the threshold cells show the minimum population share of the U-stratum (π_U, as a fraction of the total population) above which the ATT turns negative. For example, at $ATE = 0.04$ and $\lambda = 2$ (meaning that for every member of the T-stratum who is treated, two members of the U-stratum are treated, per capita), the ATT is negative whenever the U-stratum constitutes more than 4% of the population. Conversely, the top-left region applies when the ATE is negative and the T-stratum is more likely to be treated ($\lambda < 1$). At $ATE = -0.10$ and $\lambda = 1/3$ (meaning for every member of the U-stratum who is treated, three members of the T-stratum are treated, per capita), the ATT is positive whenever the T-stratum exceeds 5% of the population.

		Selection ratio $\lambda = \rho_U/\rho_T$						
		T more likely treated				U more likely treated		
		$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	1	$\frac{3}{2}$	2	3
ATE	-0.10	$\pi_T > 5\%$	$\pi_T > 10\%$	$\pi_T > 20\%$	✓	✓	✓	✓
	-0.04	$\pi_T > 2\%$	$\pi_T > 4\%$	$\pi_T > 8\%$	✓	✓	✓	✓
	-0.02	$\pi_T > 1\%$	$\pi_T > 2\%$	$\pi_T > 4\%$	✓	✓	✓	✓
	0.02	✓	✓	✓	✓	$\pi_U > 4\%$	$\pi_U > 2\%$	$\pi_U > 1\%$
	0.04	✓	✓	✓	✓	$\pi_U > 8\%$	$\pi_U > 4\%$	$\pi_U > 2\%$
	0.10	✓	✓	✓	✓	$\pi_U > 20\%$	$\pi_U > 10\%$	$\pi_U > 5\%$

Table 2: Minimum share of the countervailing stratum required for the ATT to have opposite sign from the ATE, as a function of the ATE and the selection ratio $\lambda = \rho_U/\rho_T$. When $\lambda > 1$, the U-stratum (demobilizers) is more likely to be treated than the T-stratum (mobilizers); when $\lambda < 1$, the reverse. Cells show the threshold from Corollary 3. Off-diagonal check marks indicate regions where differential selection reinforces population share differences, so the ATE and ATT always agree in sign. The $\lambda = 1$ column indicates no differential selection, so signs agree by Corollary 2. The table is symmetric around $\lambda = 1$ and $ATE = 0$.

The framework and Table 2 are entirely data-agnostic: they apply to any setting with a binary treatment and a binary outcome where treatment plausibly generates opposing responses among

distinct subpopulations and exposure to treatment covaries with response type. Any empirical researcher can use the table to assess, for a given ATE magnitude and belief about differential selection, whether the ATE and ATT might diverge in sign. In the next section, I apply this framework to criminal victimization and political participation.

2 Application to Criminal Victimization

Extant theoretical accounts of the participatory consequences of criminal victimization, drawing heavily from work on civil war and policing violence, posit victimization may participation decisions through two competing mechanisms. The first encompasses processes that increase participation: post-traumatic growth can transform victims into civic actors, anger at victimization can generate demand for political action, shared exposure can build solidarity that lowers collective action costs, or victims may acquire instrumental incentives to demand security or punish incumbents (Bateson, 2012; Blattman, 2009; Koos and Traunmüller, 2024; Gilligan, Pasquale and Samii, 2014; Ley, 2022). The second encompasses processes that suppress it: by engendering fear, crime can raise the perceived costs of public engagement, and victimization can erode trust in the state's capacity to protect (Ley, 2018; Hale, 1996). These two broad channels, and the many mechanisms they encompass, map onto the *Participate if Treated* and *Participate if Untreated* strata, respectively.

Criminological research emphasizes that a central feature of crime is that exposure is not randomly distributed. Victimization risk varies systematically with mobility, risk preferences, socioeconomic status, and access to protection. Wealthier and more risk-averse individuals tend to be better able or more willing to avoid victimization (Skogan, 1995; Hale, 1996), whereas more mobile and risk-prone individuals embedded in their communities face greater exposure and often weaker protection (Rader, 2004; Browning, Pinchak and Calder, 2021; Boggs, 1965; Guedes, Domingos and Cardoso, 2018). These same characteristics are also correlates of political participation in the literature, with mobility, risk tolerance, and community embeddedness associated

with higher rates of civic and political engagement (Gerber et al., 2011; Kam, 2012). Crime is therefore a setting in which who is exposed and how they respond are plausibly linked.

Much of the empirical literature concludes that victimization, on average, increases non-electoral civic and political engagement (Bateson, 2012; Dorff, 2017; Boulding, Mullenax and Schauer, 2022; Ley, 2022). This evidence is largely generated by research designs that target population-average or conditional average treatment effects. Bateson (2012), for example, conditions on observed covariates and invokes conditional ignorability, estimating how participation would change if victimization were randomly assigned within covariate strata. This answers a coherent counterfactual question, but it does not describe how participation changes among those who are actually victimized when exposure is selective.[5] If subpopulations facing higher victimization risk are also more likely to be demobilized by victimization, population-average effects need not describe effects among victims.

2.1 Data

I focus on Latin America, where victimization is widespread (Vilalta, 2020; LAPOP, 2022) and where much of the research on crime and participation has been conducted (Visconti, 2020; Ley, 2018; Blanco, 2013; Marshall, 2022).

As the primary data source, I employ an extended version of the LAPOP survey, which Bateson (2012) uses for their Latin American results. I pool all surveys conducted in 20 Latin American countries between 2010 and 2019, yielding approximately 150,000 respondents from 94 survey waves. Each survey round was designed to represent the voting-age population of the country in that year.[6] LAPOP measures self-reported victimization in the past 12 months and several forms of civic and political engagement. I focus on five outcomes: protest participation, political meeting attendance, community meeting attendance, participation in community problem-solving

[5] This distinction is not a critique of observational identification. Confounding concerns whether a given estimand is estimated without bias; here, different research designs can target different causal quantities for the same treatment and outcome.

[6] See Appendix A2.1.1 for sampling details.

activities, and willingness to vote in upcoming national elections if they were held next week.

2.2 Empirical Strategy

I implement the partial identification procedure using the LAPOP data. From the pooled data, I can retrieve the proportion of respondents who report being victims, $Pr[Z = 1]$ and its converse probability, the proportion of respondents who report engaging in each of the participatory activities, $Pr[Y = 1]$, and its converse, and the proportion of victimized/unvictimized respondents who report participating $Pr[Y = 1|Z = 1]$, $Pr[Y = 1|Z = 0]$, respectively. Table 3 reports the observed probabilities in the data for each of the outcomes.[7]

Leveraging the structure of the principal-strata framework, these observed quantities restrict the possible values of the population shares of each stratum (π_θ) and their corresponding treatment probabilities (ρ_θ). The expressions derived in Section 1 imply that any feasible parameter vector must reproduce the observed participation moments through the identities linking $(\pi_\theta, \rho_\theta)$ to $(Pr[Y = 1], Pr[Y = 1 \mid Z = 1], Pr[Y = 1 \mid Z = 0], Pr[Z = 1])$. For each feasible vector, I then compute the ATE and ATT directly using the closed-form expressions established earlier:

$$ATE = \pi_T - \pi_U, \qquad ATT = \frac{\pi_T \rho_T - \pi_U \rho_U}{Pr(Z = 1)}.$$

To characterize the full set of parameter values consistent with the data, I explore the feasible parameter space for $\{\pi_\theta, \rho_\theta\}$ for all $\theta \in \{A, T, U, N\}$. The approach follows the logic of partial identification: instead of selecting assumptions that point-identify either estimand, I recover the entire identified set of parameter combinations implied jointly by the observed moments and the model's structure (Kline and Tamer, 2023). This proceeds as follows.

1. I construct a fine grid over all possible population shares $(\pi_A, \pi_T, \pi_U, \pi_N)$ such that $\sum_\theta \pi_\theta = 1$.

[7] These proportions are estimated from survey data and therefore contain uncertainty. However, the large sample size yields very precise estimates, rendering the incorporation of this uncertainty into subsequent simulations negligible and immaterial to the results. For simplicity and clarity, I omit this uncertainty from the explanation.

2. For each candidate vector of population shares, I use the observed moments in Table 3 to solve for the implied treatment probabilities ρ_T and ρ_U that rationalize the observed conditional participation rates.

3. I then solve for ρ_A and ρ_N using the marginal probability of victimization, $Pr(Z = 1)$.

4. Any parameter vector that violates logical or probabilistic constraints (e.g., treatment probabilities outside $[0, 1]$ or participation rates inconsistent with potential outcomes) is discarded.

5. For every remaining feasible parameter vector, I compute the corresponding ATE and ATT using the expressions above, as well as the implied differential selection ratio $\lambda = \rho_U/\rho_T$ and stratum-size ratio $\eta = \pi_T/\pi_U$, which jointly govern whether the ATT agrees in sign with the ATE (Section 5). Because the ATT depends on heterogeneous treatment probabilities, while the ATE depends only on population shares (Proposition 1), differences between the two estimands across feasible vectors reveal the extent to which heterogeneous exposure patterns can reverse or attenuate conclusions about victimization's political effects.

This procedure yields a distribution of ATEs and ATTs, the identified set, that fully characterizes the causal effects consistent with the LAPOP data under heterogeneous treatment probabilities. The contrast between these identified sets illustrates the empirical relevance of the theoretical results in Section 1: whenever individuals who reduce participation after victimization are more likely to be victimized (i.e., $\rho_U > \rho_T$), the ATE and ATT diverge, often in magnitude and sometimes in sign. All algebraic derivations, feasibility constraints, and computational details are reported in Appendix C.

With this set of parameters, I can compare the realized change in outcomes under actual exposure patterns to the hypothetical change under random exposure. I allow for the effect of victimization to mobilize or demobilize participation. That is, I assume $\pi_U \geq 0$.[8]

[8]However, Figure A4 in the Appendix reports the results when the effect of victimization on participation is assumed to be only increasing.

Participation Type	N	$Pr(Z=1)$	$Pr(Y=1)$	$Pr(Y=1\|Z=1)$	$Pr(Y=1\|Z=0)$
Attended Protest	149,728	0.21 (0.001)	0.08 (0.0007)	0.14 (0.002)	0.07 (0.0007)
Community Problem-Solving	94,022	0.19 (0.001)	0.36 (0.001)	0.43 (0.004)	0.34 (0.002)
Attended Community Meeting	149,405	0.21 (0.001)	0.29 (0.001)	0.32 (0.003)	0.28 (0.001)
Attended Political Meeting	137,308	0.20 (0.001)	0.16 (0.0009)	0.18 (0.002)	0.15 (0.001)
Would vote next week	127,256	0.22 (0.001)	0.81 (0.001)	0.85 (0.002)	0.80 (0.001)

Table 3: Descriptive statistics from the LAPOP survey for each measure of political participation. N: respondents with non-missing participation and victimization responses. $Pr(Z = 1)$: victimization rate. $Pr(Y = 1)$: unconditional participation rate. $Pr(Y = 1 \mid Z = 1)$ and $Pr(Y = 1 \mid Z = 0)$: participation rates among victims and non-victims, respectively. Standard errors in parentheses.

3 Results: Crime Victimization

After repeating the algorithm detailed in the previous section for each of the participatory outcomes, I obtained 12.1 million, 16 million, 20.6 million, 18.34 million, and 3.63 million feasible parameter vectors $\{\pi_\theta, \rho_\theta\}$ $\forall\theta \in \{A, T, U, N\}$, each implying a specific λ and η, for the protest attendance, political meeting attendance, community meeting attendance, community problem-solving outcomes, and willingness to vote respectively. These are an internally consistent set of values that bound the parameter's actual value. The difference in the sizes of the parameter sets depends on how informative the data is; for infrequent outcomes, fewer parameter sets are consistent with the data.

For each set, I can then calculate the corresponding ATE and ATT of victimization on each of the measures of political participation. Figure 1 plots the results. Each point in the graph corresponds to a plausible combination of π_θ, ρ_θ for a given participatory outcome. The red diagonal line that passes through the origin indicates the points where the ATE and the ATT are equal. For all the points below, the ATE is larger than the ATT. Conversely, the ATT is larger than the ATE for all points above that line. The color of the point indicates the difference between the probability of becoming a victim for individuals who become mobilized due to victimization and the probability of becoming a victim for individuals who respond by decreasing political participation in the set.

The implied ATE is almost always larger than the corresponding ATT across all measures

of non-electoral political participation. When participating in a protest is the outcome, the ATE is larger than the ATT in 99% of the parameter sets. When attending political meetings is the outcome; the same holds for 96% of the parameter sets. When it is attending community meetings, the ATE is larger in 83% of the parameter sets. When participating in community problem-solving activities is the outcome, the percentage equals 64%. Further, we can see that the points above the red diagonal line correspond to the largest differences between $\rho_T - \rho_U$. For the ATT to be larger than the ATE, selection into victimization has to be strongly correlated with the treatment, making individuals more participative.

While the majority of the ATEs compatible with the data are positive for all four measures of non-electoral participation (85% for protest attendance, 79% for political meeting attendance, 67% for community meeting attendance, and 63% for community problem solving), the converse is true for the ATTs across measures. Specifically, only 9% of the ATTs are positive for protest attendance, only 13% for political meeting attendance, 29% for community meeting attendance, and 46% for community problem-solving. Even for the most frequently reported outcome, community meeting attendance, in most cases, the ATTs compatible with the data are negative. As reported in Table 3, the larger share of positive and ATTs reflects the larger conditional probabilities of participating in these activities given victimization implied in the respondents' answers.

The results from the willingness-to-vote question, if elections were held the following week, show that electoral participation, to the degree that this is a construct-valid question, shows a markedly different pattern than non-electoral forms of engagement. While the ATE exceeds the ATT across most non-electoral participation measures, this relationship reverses for voting intention. Here, the ATE exceeds the ATT in only 1% of parameter sets. More strikingly, in 89% of parameter sets compatible with the data, the ATE of victimization on willingness to vote is *negative*, while in 95% of the sets, the ATT is positive.

Three considerations help explain this reversal. First, voting may be relatively less costly than non-electoral participation. For the marginal individual, it requires no collective organization,

no sustained community presence, and no resources beyond showing up. The mechanisms that suppress costly participation after victimization may therefore not suppress voting. Second, strata may be outcome-specific. An individual can be in the *Participate if Untreated* stratum for protest but in the *Participate if Treated* stratum for voting. With roughly 8% of respondents reporting willingness to vote and 20% victimization, the population is dominated by *Always* voters for whom voting behavior is unaffected by victimization, consistent with the empirical stickiness of electoral participation (Coppock and Green, 2016). Third, the reversal is consistent with extant evidence. Sønderskov et al. (2022), using Danish panel data, find that victimization increases turnout by 2 to 3 percentage points within individuals but that the between-individual association is large and negative. Their finding also provides direct evidence that exposure probabilities differ across response types: individuals who would increase turnout after victimization appear less likely to be victimized in the first place.

What drives the divergence between the ATE and ATT: the relative size of the mobilized and demobilized strata, or their differential exposure to victimization? Figure 2 plots the cumulative distribution of these two quantities across all parameter vectors compatible with the LAPOP data. The left panel shows $\eta = \pi_T/\pi_U$, the ratio of mobilizers to demobilizers; the right panel shows $\lambda = \rho_U/\rho_T$, the ratio of demobilizer to mobilizer victimization risk. Both panels use a log scale, with the dashed line marking the reference value of 1.

The left panel shows that across all four non-electoral outcomes, the vast majority of the identified set has $\eta > 1$, with mobilizers outnumbering demobilizers. Protest has the highest η values, with a median around 3.5; the community outcomes cluster lower, with medians between 1.7 and 2.6. The cumulative proportion at which each curve crosses the dashed line equals the share of vectors with $ATE > 0$, since $ATE = \pi_T - \pi_U$. For voting, the pattern reverses: $\eta < 1$ for most of the identified set, indicating that demobilizers outnumber mobilizers and that the ATE is predominantly negative.

The right panel tells the complementary story. For all four non-electoral outcomes, $\lambda > 1$

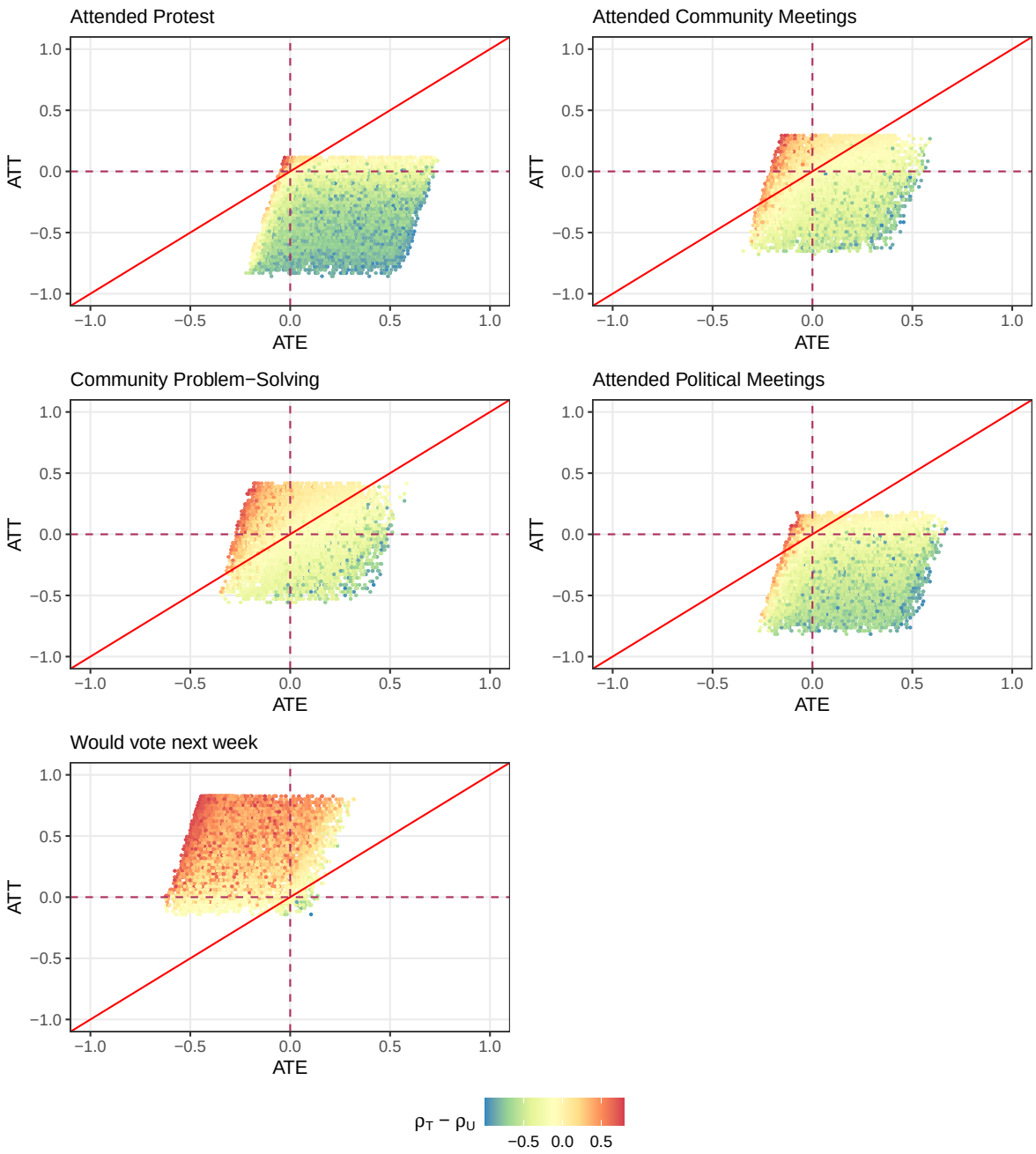

Figure 1: Figure shows the combination of potential ATTs and ATEs of victimization on four measures of political participation, conditioning on observed participation, and criminal victimization (LAPOP, 2022).

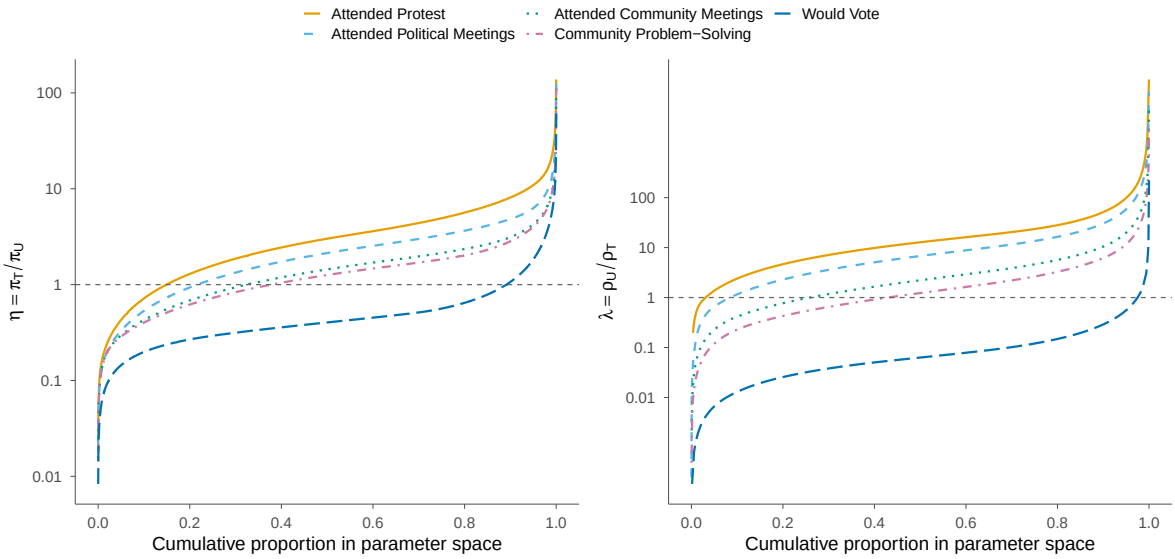

Figure 2: Cumulative distribution of the stratum-size ratio $\eta = \pi_T/\pi_U$ (left) and the differential selection ratio $\lambda = \rho_U/\rho_T$ (right) across parameter vectors compatible with the LAPOP data. Log scale. The dashed line marks 1: above it, mobilizers outnumber demobilizers (left) or demobilizers face higher victimization risk (right). The cumulative proportion at which a curve crosses 1 equals the share of the identified set with $ATE > 0$ (left) or $\rho_U > \rho_T$ (right).

across the vast majority of the identified set: demobilizers face higher victimization risk than mobilizers. Protest and political meetings show the strongest pattern, with λ rising above 1 early in the cumulative distribution. Voting again reverses: $\lambda < 1$ for most vectors, meaning mobilizers are more likely to be victimized than demobilizers.

The results suggest that if victimization were random, it would likely increase non-electoral participation, consistent with Bateson (2012) and the field's prevailing understanding.[9] However, because individuals who would participate less after victimization are disproportionately likely to become victims, the identified sets suggest that crime reduces all four types of non-electoral participation in the region. The pattern reverses for willingness to vote: crime, were it random, would likely reduce willingness to vote, but given actual patterns of victimization, it likely increases it.

[9]Results in Bateson (2012) are based on one wave of the LAPOP data. Section A2.3 in the Appendix replicates the results from Bateson (2012) with repeated cross-sections of the LAPOP data, as well as within-country analyses.

		ATE			ATT			ΔY	
Participation Type	$Pr(Z=1)$	**Min**	**Max**	$Pr(ATE<0)$	**Min**	**Max**	$Pr(ATT<0)$	**Min**	**Max**
Attended Protest	0.19	-0.23	0.76	0.15	-0.84	0.12	0.91	-0.18	0.02
Community Problem-Solving	0.19	-0.38	0.61	0.37	-0.57	0.42	0.53	-0.11	0.08
Attended Political Meetings	0.20	-0.28	0.70	0.21	-0.82	0.16	0.87	-0.17	0.03
Attended Community Meetings	0.21	-0.36	0.63	0.32	-0.68	0.30	0.71	-0.14	0.06
Would vote next week	0.22	-0.65	0.33	0.89	-0.13	0.84	0.05	-0.03	0.18

Table 4: Identified sets of ATE, ATT, and $\Delta Y = ATT \times Pr(Z=1)$ from all feasible parameter vectors compatible with the LAPOP data, without constraining ρ_U or ρ_T. Min and Max report the bounds of each identified set. $Pr(ATE<0)$ and $Pr(ATT<0)$ report the share of feasible parameter vectors for which each estimand is negative.

What are the magnitudes of the changes in participation? Recall from expression (1) that the realized change in population outcomes equals the ATT scaled by the probability of treatment. Table 4 synthesizes all the results. Participation in protests could decrease by as much as 18 pp, engagement in community problem-solving could decrease by 11 pp, political meeting attendance could decrease by 17 pp, while community meeting attendance could decrease by as much as 14 pp. Willingness to vote next week, however, at worst could decrease by 3 pp. Conversely, conditional on the data, protest attendance could *at best* increase only by 2 pp; community problem solving by 8 pp; political meeting attendance by 3 pp; and community meeting attendance by 6 pp. Hypothetical williness to vote could increase by at most 18 pp. The *upper bound effects* within the identified set of non-electoral participation estimands are comparable in magnitude to, and often smaller than, the ATEs identified in Bateson (2012).

One concern is that these results reflect peculiarities of the LAPOP data. For example, we might be worried that LAPOP over-represents urban areas, under-represents dangerous areas, or yields survey responses that do not generalize. To address such concerns, I analyze data from Dorff (2017), who conducted a nationally representative survey of 1,000 Mexican respondents that included questions about victimization and both electoral and non-electoral participation. I apply the same partial identification procedure from the main paper to this alternative dataset. I

present the results in Table A6 in the Appendix. I find strikingly similar results that point in the same direction: the identified sets suggest that while the ATE is likely positive, the ATT is almost certainly smaller and likely negative.

4 Panel Evidence from Mexico

The results from the previous section give us a distribution of the frequency, magnitude, and direction of plausible causal effects compatible with the LAPOP data. In this section, I analyze a panel survey to provide evidence of who becomes a victim in a real-world setting and how victims react to victimization, moving from the plausible to the empirical.

I use a quarterly rotating panel survey of Mexican city residents that asks questions related to crime and insecurity, the National Survey of Urban Public Safety, ENSU, for its name in Spanish (INEGI, 2024). This survey has been conducted four times per year since 2013[10] and, starting in 2017, has included direct questions regarding personal victimization. This rolling survey panel allows me to observe individuals at five different points over 15 months. In each quarter, respondents are asked about experiences with the state, crime, and conflict, and in at least two waves per year, they are asked about direct victimization. I keep all survey waves from 2017, when crime victimization was first incorporated, through the last quarter of 2023, resulting in 560k responses from 174K unique respondents collected over 27 quarters.

Unlike the LAPOP data, the panel structure of ENSU allows me to provide direct evidence on both conditions that the framework requires. First, by comparing future victims with non-victims before any victimization occurs, I assess whether exposure is selective: if it is, future victims should differ systematically from their neighbors even before becoming victims. Second, by exploiting within-individual variation, I estimate how victimization changes attitudes and behavior among those who experience it, providing evidence on the direction of response among the treated.[11]

[10] Except the second quarter of 2020 due to the COVID emergency.

[11] Like any survey measure, self-reported victimization may be subject to heterogeneous reporting: individuals may

To assess whether exposure to victimization is selective, I identify ENSU questions that map onto the characteristics the framework calls for to vary across strata: mobility patterns, exposure to criminal and violent behavior, feelings of unsafety, and interpersonal conflicts. Higher mobility increases exposure to crime (Rader, 2004; Browning, Pinchak and Calder, 2021); witnessing criminal behavior directly indexes victimization risk; and fear of crime correlates with perceived vulnerability (Guedes, Domingos and Cardoso, 2018). Critically, these same characteristics are also correlates of political participation: extroverted, mobile, community-oriented, and risk-prone individuals participate more (Gerber et al., 2011; Kam, 2012; Ley, 2022). While ENSU does not directly measure political participation, these behavioral and attitudinal measures serve as proxies for the participatory profiles that distinguish strata in the framework.

To assess reactions to victimization, I estimate how victimization changes these same attitudes and behaviors among those who experience it. I exploit within-individual variation using two estimators: a two-way fixed effects estimator and the fixed-effect counterfactual estimator proposed by Liu, Wang and Xu (2022). I focus on measures of crime prevention behavior, fear of crime, trust in law enforcement, and mobility. If victimization demobilizes, victims should become less mobile, more fearful, and less trusting; the behavioral profile of the *Participate if Untreated* stratum.

4.1 Exposure to Crime

I subset the data and retain only respondents who never reported criminal victimization and respondents who reported criminal victimization in some, but not all, survey waves. For the latter, I retain responses only before their first reported instance of victimization. Consequently, I compare the responses of this group of future victims *before they reported victimization* with those of their never-victimized neighbors—either from the same sampling unit or the same locality—in the same quarter.

interpret events as criminal or non-criminal differently, and such sensitivity could correlate with political attitudes and behavior (Skogan, 1982; Boulding, Mullenax and Schauer, 2022). Two descriptive analyses in Appendix Appendix E show that self-reported victimization tracks municipal-level criminal victimization rates as expected.

I estimate

$$PM_{i[q,g]} = \beta_1 * FutureVictim_{i[q,g]} + \theta_{q\times g} + \epsilon_{i[q,g]} \tag{3}$$

where $PM_{i[q,g]}$ is respondent $i's$ self-reported behavior, experiences, or attitude in year-quarter q, and neighborhood/census tract or locality g, or a binary measure that takes the value of 1 if respondent i reports that type of behavior or experience and 0 otherwise. $FutureVictim_{i[q,g]}$ is a binary variable that takes the value of 1 when respondent i will report having been a victim of crime in a future survey and 0 otherwise. $\theta_{q\times g}$ represent either $Neighborhood \times Quarter$ fixed effects, or $Locality \times Quarter$ fixed effects. $\epsilon_{i[q,g]}$ are robust errors clustered at the primary sampling unit.

Figure 3 shows results. When we compare future victims with never-victims living in the same locality in that same quarter (dark blue) or the same small neighborhood (light blue), future victims report having more conflicts, including conflicts that resulted in physical violence, conflicts with authorities, conflicts with the police, and conflicts with strangers on the street, *before* being victimized. Future victims also witnessed gangs, robberies, and vandalism, and heard gunshots more frequently. They also report feeling unsafe in the city, on the streets, on public transport, and even in their house. Last, they report leaving their home more frequently.

Overall, the results link future victimization with increased exposure to crime and show that victims and non-victims were differentially exposed to crime even before becoming victims. These findings are consistent with future victims disproportionately resembling the *Participate if Untreated* stratum: before victimization, they are more mobile, more exposed to criminal activity, and more embedded in conflictual social interactions, which matches the behavioral profile associated with higher political participation in extant research (Gerber et al., 2011; Kam, 2012).

4.2 Reactions to Crime

I next estimate how victimization changes attitudes and behavior among those who experience it.

$$PM_{i[q]} = \beta_1 * Victim_{i[q]} + \theta_i + \gamma_q + \epsilon_{i[q]} \tag{4}$$

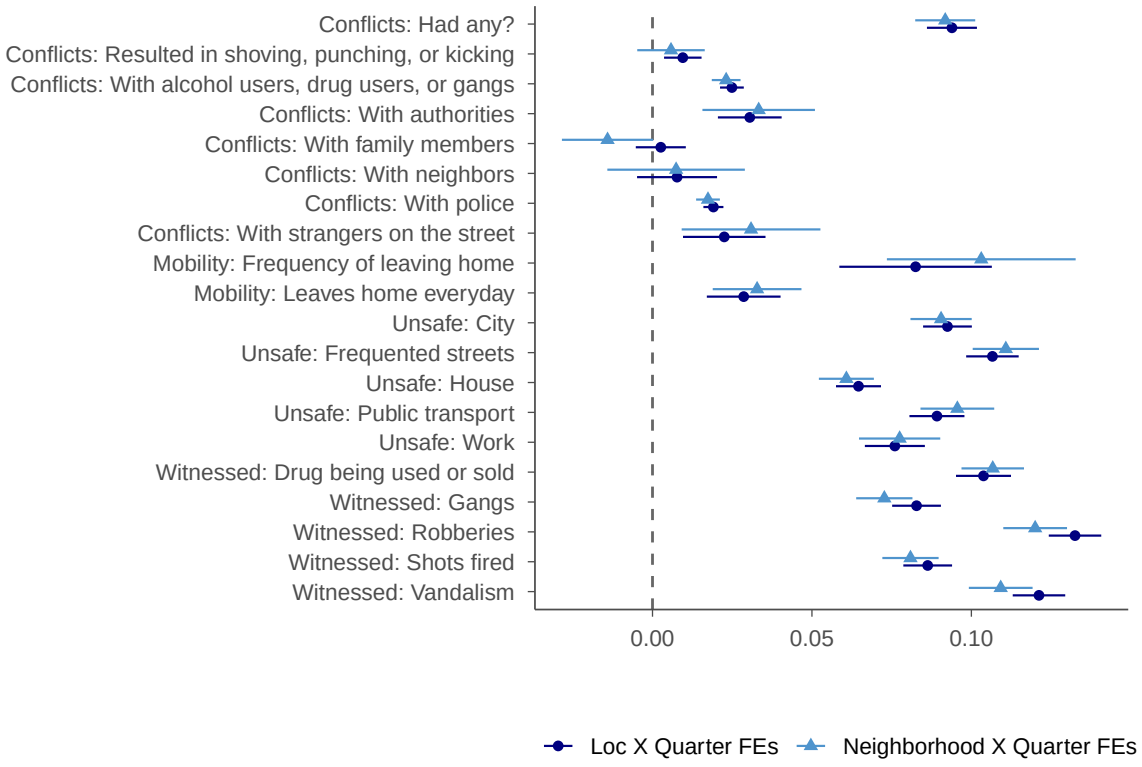

Figure 3: Cross-sectional differences between future victims and non-victims living in the same locality (dark blue) and the same neighborhood/census tract (light blue) on self-reported outcomes relating to exposure to crime. Robust standard errors clustered at the primary sampling unit. Full regression results in Table A12.

where $PM_{i[q]}$ is either a continuous measure of respondent $i's$ self-reported behavior, experiences, or attitude in year-quarter q, and neighborhood or locality g, or a binary measure that takes the value of 1 if respondent i reports that type of behavior, experience or attitude and 0 otherwise. $Victim_{i[q]}$ is an indicator variable that takes the value of 1 when respondent i reports having been victimized in the past six months in quarter q and 0 otherwise. Because victimization is asked every six months, while response variables are asked every three months, each respondent's victimization report is extended to cover the prior period. θ_i are individual fixed effects while γ_q are period fixed effects. $\epsilon_{i[q]}$ are robust errors clustered at the primary sampling unit. I use a TWFE estimator of the ATT as well as the counterfactual estimator with equal weights proposed by Liu, Wang and Xu (2022).

Figure 4 shows the results. It plots the ATT of victimization on several behavioral and attitudinal outcomes, per specification (4). Results show victimization changed attitudes and behaviors in a way that is consistent with individuals becoming demobilized after becoming victims, regardless of the preferred estimator.

Compared with their behavior before becoming crime victims, respondents report taking more steps to prevent crime, including leaving home less frequently and visiting friends and family less often. Along the same line, they report being less mobile after victimization and are less likely to say they leave home every day. These two findings suggest that victims become less extroverted. Further, victims report trusting law enforcement authorities less, being more afraid of crime, and taking more steps to prevent victimization. These latter results suggest a reduction in people's risk acceptance and the adoption of more constrained behavior, consistent with what we would expect from the *Participate if Untreated* stratum. The participatory consequences of such behavior change should unambiguously lead to less participation generally, including in political protests and community organizing specifically (Gerber et al., 2011; Kam, 2012; Ley, 2022; Skogan, 1986; Ley, 2018). These findings further indicated that victims could be conceptualized as mostly drawn from the *Participate if Untreated* stratum.

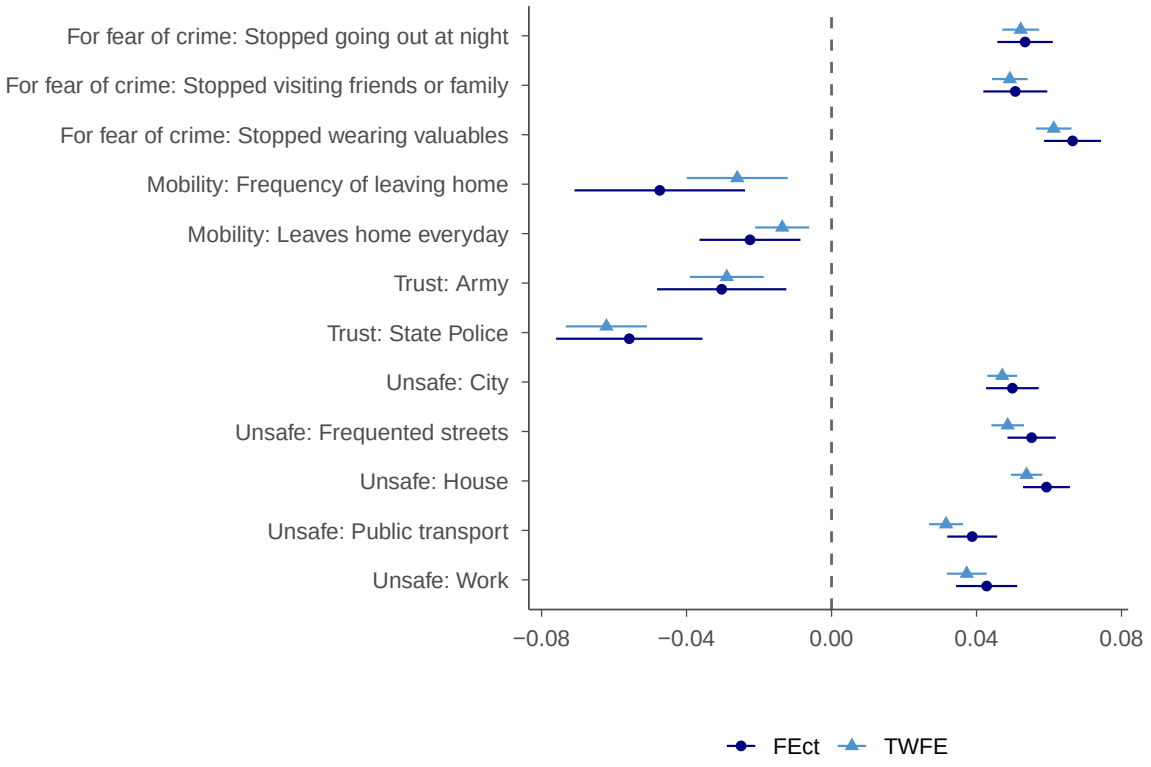

Figure 4: Within-individual effects of victimization on self-reported behavioral and attitudinal measures, estimated with the fixed-effect counterfactual estimator from Liu, Wang and Xu (2022) (dark blue) and a two-way fixed effects estimator (light blue). Robust standard errors clustered at the primary sampling unit. Full regression results in Table A13.

Overall, the findings from the Mexico survey show that the risk of victimization is heterogeneous and higher for individuals who become demobilized after becoming victims, consistent with the results from the prior LAPOP analysis.

5 Disciplining the Identified Set with Panel Evidence

The LAPOP cross-section established that the majority of the parameter space favors demobilization among the victimized. The ENSU panel confirmed that demobilizers are more likely to be victimized and that victimization demobilizes within individuals. In this section, I use what the ENSU panel establishes about the direction of selective exposure to assess whether the ATT can be signed.

As established in Section 1, whether the ATT agrees in sign with the ATE depends on two quantities:

1. The *differential selection ratio*, $\lambda = \rho_U/\rho_T$: how much more likely demobilizers are to be victimized compared to mobilizers. When $\lambda = 1$, both strata face identical victimization probabilities and sign agreement is guaranteed (Corollary 2). When $\lambda > 1$, demobilizers are disproportionately exposed to treatment.

2. The *stratum-size ratio*, $\eta = \pi_T/\pi_U$: how numerous mobilizers are relative to demobilizers. When $\eta > 1$, mobilizers outnumber demobilizers in the population.

When $ATE > 0$, Proposition 3 establishes that the ATT is negative if and only if $\lambda > \eta$. In words: the ATT reverses sign when the overexposure of demobilizers exceeds their relative scarcity. This is a joint condition on two parameters, and signing the ATT requires establishing both: that demobilizers are more exposed ($\lambda > 1$), and that they are not too scarce relative to mobilizers (η not too large). I address each in turn.

The LAPOP cross-section alone cannot establish that $\lambda > 1$: the identified set includes parameter vectors on both sides of unity. But the evidence points strongly in one direction. The right

panel of Figure 2 shows that for all four non-electoral outcomes, $\rho_U > \rho_T$ for the vast majority of feasible vectors—demobilizers face higher victimization risk than mobilizers. The ENSU panel establishes the direction of λ: future victims are more exposed before victimization, and victimization demobilizes within individuals, jointly implying $\lambda > 1$. Imposing this constraint from the panel data, the question of whether the ATT is signed reduces to a single parameter: how large is η? In the next table, I progressively tighten an upper bound on η to assess how extreme the stratum-size ratio must be for the ATT to remain positive.

Given $\lambda > 1$, the remaining question is whether η is small enough for the sign reversal to bite. Recall that $\eta = \pi_T/\pi_U$ counts how many mobilizers there are per demobilizer: $\eta = 2$ means that for every person whom victimization demobilizes, two are mobilized. When η is large, the mobilizer stratum dilutes the overexposure of demobilizers, and the ATT can remain positive despite $\lambda > 1$. This is the barrier to signing.

The LAPOP identified set tells us where η falls. Conditional on $ATE > 0$, the median η across outcomes ranges from 1.7 to 3.5; the 10th percentile ranges from 1.1 to 1.5. In every case $\eta > 1$: mobilizers outnumber demobilizers. But how large should we *expect* η to be? Since $ATE = \pi_T - \pi_U$, straightforward algebra gives $\eta = 1 + ATE/\pi_U$. If the protest ATE is approximately 4 percentage points (Bateson, 2012) and the demobilizer stratum is roughly 12% of the population (the LAPOP median π_U for protest), then $\eta \approx 1.33$. For community outcomes, with smaller ATEs and larger π_U (≈ 0.22–0.25), η falls even closer to 1. Published ATE estimates and the LAPOP-implied stratum sizes jointly place η in the range of 1 to 1.5—well below the medians of the unrestricted identified set.

Table 5 puts these pieces together for non-electoral outcomes, where the literature finds $ATE > 0$. I fix $\lambda > 1$, as established by the ENSU panel evidence, and progressively tighten an upper bound on η, the number of mobilizers per demobilizer. For each restriction, among LAPOP parameter vectors satisfying $ATE > 0$ and the imposed bound, I compute the share that yield $ATT < 0$. When the share reaches 100%, the ATT is *signed*: every feasible vector agrees on its

direction. A corresponding analysis for voting, where the sign-flip runs in the opposite direction, is reported in the Appendix.

	Bound on $\eta = \pi_T/\pi_U$ (mobilizers per demobilizer)				
	Unrestricted	≤ 2	≤ 1.5	≤ 1.1	≤ 1.04
Share of feasible vectors with $ATT < 0$, given $ATE > 0$ and $\lambda > 1$.					
Protest	0.90	0.98	0.99	>0.99	**Signed** $-$
Community Problem-Solving	0.71	0.80	0.87	0.97	>0.99
Community Meetings	0.79	0.89	0.93	0.97	0.98
Political Meetings	0.89	0.97	0.98	>0.99	**Signed** $-$

Table 5: Sensitivity of ATT sign identification to restrictions on the stratum-size ratio. Cells report the share of LAPOP-compatible parameter vectors with $ATT < 0$, conditional on $ATE > 0$ and $\lambda > 1$ (demobilizers more exposed). Columns impose progressively tighter upper bounds on $\eta = \pi_T/\pi_U$. **Signed** $-$: 100% of feasible vectors yield $ATT < 0$.

In table 5 the first column imposes only $\lambda > 1$, with no restriction on how numerous mobilizers are. Even so, 71–90% of non-electoral parameter vectors yield $ATT < 0$—the ATT is already negative for the vast majority of the identified set before imposing any bound on η. Moving rightward, at $\eta \leq 2$ (at most two mobilizers per demobilizer) the shares reach 80–98%. At $\eta \leq 1.5$ (three mobilizers for every two demobilizers), all four outcomes exceed 87%. At $\eta \leq 1.1$ (mobilizers barely outnumber demobilizers), protest and political meetings exceed 99% and the community outcomes reach 97%. At $\eta \leq 1.04$—just 104 mobilizers per 100 demobilizers—protest and political meetings are signed negative, and the community outcomes reach 98% and above.

Where do published ATEs place us? The Bateson-implied $\eta \approx 1.33$ for protest falls between the third and fourth columns, squarely in the range where 87–99% of vectors give $ATT < 0$. The signing threshold of $\eta \leq 1.04$ corresponds to $ATE \leq 0.04 \times \pi_U$—less than half a percentage point for protest—far below any published estimate. The ATT is negative for the great majority

of the identified set at empirically plausible values of η; formal signing requires stratum sizes that imply an implausibly small ATE. The π_U threshold from Equation (2) reinforces this: at a modest $\lambda = 1.4$ and $ATE \approx 0.04$ for protest (Bateson, 2012), the ATT reverses sign whenever the demobilizer stratum exceeds 10% of the population. The LAPOP-implied median π_U for non-electoral outcomes ranges from 0.12 (protest) to 0.25 (community problem-solving), well above this threshold.

In sum, for non-electoral participation, where the ATE is positive, the overexposure of demobilizers reverses the ATT to negative for the great majority of feasible parameter vectors. Formal sign identification requires only that mobilizers do not vastly outnumber demobilizers—a condition well within the range implied by published ATE estimates.

6 Applications to other Forms of Violence

Criminal victimization is one setting where the probability of exposure and the effect of treatment plausibly covary, but it is not unique. The confluence of selective exposure and heterogeneous responses is a general feature of many forms of violence studied in political science. To demonstrate that the framework applies beyond crime, I turn to two related settings where exposure is plausibly selective and theoretical accounts support heterogeneous responses: civil war abduction in Uganda and police contact in the United States.

6.1 Civil War Violence

Studies of civil war violence have generally found that exposure makes victims more likely to participate in politics, attributing this to post-traumatic growth and increased prosociality (Bellows and Miguel, 2009; Blattman, 2009; Voors et al., 2012). One influential paper is Blattman (2009), which examines how forced recruitment into the LRA insurgency in Northern Uganda shaped downstream political participation. Leveraging theorized quasi-random variation in abduction, he estimates the ATE and finds that abduction significantly increased voting and doubled the likelihood of becoming a community leader.

I apply the partial identification procedure to data from Blattman (2009) to derive the identified sets of ATEs, ATTs, and ρs. Detailed results are reported in Section A6.1. For electoral participation, 77% of the ATEs in the identified set are positive, and 75% of the ATTs are as well — unlike the LAPOP results, the two estimands largely agree. For non-electoral participation, however, the pattern mirrors the crime application. Only 35% of the identified ATEs imply that abduction increased community leadership, despite the original point estimate being positive and significant. And between 93%–96% of the identified ATTs are negative. If abduction were random, it would likely increase voting and some forms of civic engagement. If it were selectively experienced, it likely depressed non-electoral participation by a substantial margin.

If abduction was random, as theorized by the author, then by Corollary 1, the ATE equals the ATT, and research designs targeting either estimand are equally suitable for measuring actual changes in participation. The partial identification exercise shows what is at stake when that assumption fails: if exposure to abduction was even moderately selective, the positive effects on non-electoral participation that the ATE implies may not describe what happened to those who were actually abducted. Whether it was or not is a theoretical assumption that requires theoretical justification.

6.2 Contact with the Police

Research on the political consequences of police contact presents competing findings. Some work documents that contact teaches individuals they are politically marginal, suppressing engagement (Weaver and Lerman, 2010; Soss and Weaver, 2017). Others find that police contact, whether direct or indirect, stimulates participation (Walker, 2014, 2020). That police engage in statistical discrimination is well established (Ba et al., 2021; Mummolo, 2017), making this a setting where exposure is plausibly selective.

Walker (2020) finds that direct and indirect police contact is associated with higher rates of both electoral and non-electoral participation, targeting the ATE as the primary estimand. I apply

the partial identification procedure to one of the data sources used, the National Crime and Politics Survey, a nationally representative U.S. survey conducted in 2013. I focus on direct police contact and use all available measures of electoral and non-electoral participation. Results are reported in Table A11.

The pattern mirrors the crime application. While most parameter sets show positive ATEs for non-electoral participation, the ATTs are predominantly negative. For voting, the pattern again reverses: 89% of parameter sets yield negative ATEs, while only 7% show negative ATTs. If police contact were random, it would increase non-electoral participation. If it is selectively experienced in ways that coincide with heterogeneity in reactions, as the abundant literature suggests it is, it likely depresses non-electoral engagement among those most frequently stopped while increasing their electoral participation.

7 Guidance for Researchers

The results in this paper reverse substantive conclusions across three distinct settings. In this section, I distill the framework and empirical findings into a checklist for researchers looking to assess whether their conclusions depend on the choice of estimand, or to design strategies that account for the coincidence of selective exposure and heterogeneous responses. The first three points apply broadly to any setting where exposure is plausibly selective. The last three are for implementing the partial identification procedure, which requires binary treatment and outcome.

1. What estimand does the analysis target? The estimand links theory to empirics by specifying the population over which a causal effect is defined and, therefore, the counterfactual comparison it supports. Researchers should understand which population their estimate is defined over and why that population is the relevant one for the claim.

2. Does the estimand incorporate selective exposure in a manner consistent with the theory? In specifying the population, the estimand also determines how heterogeneous treatment probabilities are incorporated into the empirical analysis. The ATE weights response types by

their population shares; the ATT weights them by their exposure probabilities. Both are useful quantities. The ATE answers how treatment would change outcomes if exposure were equal across subpopulations. The ATT answers how treatment changes outcomes given who is actually exposed. When these diverge, each supports a different substantive claim. Researchers should be explicit about which question they are asking and whether their estimand answers it.

3. Are the findings you compare defined for the same population? When engaging with the empirical literature, researchers should consider whether the studies they compare define effects for the same population. In the presence of selective exposure, apparent disagreement across studies may reflect differences in the estimands being targeted rather than in mechanisms or contexts. Making the estimand explicit clarifies whether findings are commensurable, which literature the study contributes to, and what the prior expectation for effects should be.

4. If treatment and outcome are binary, characterize what the data allow. The partial identification procedure requires four quantities: $Pr(Z = 1)$, $Pr(Y = 1)$, $Pr(Y = 1|Z = 1)$, and $Pr(Y = 1|Z = 0)$. From these, the researcher can characterize the identified set of ATEs and ATTs consistent with the observed moments. Any treatment and outcome can be made binary — continuous outcomes can be dichotomized at a meaningful threshold, and ordinal scales can be collapsed — though this comes at some loss of information. The binary-binary setting is thus widely applicable. Computational details and replication code are provided in Appendix C.

5. Interpret the identified set. For each estimand, check whether all feasible parameter vectors yield the same sign. Table 2 provides a quick diagnostic: given an estimated ATE and a belief about how selective exposure is, researchers can assess whether sign reversal is possible. If yes, the sign is identified. If both the ATE and ATT are sign-identified and agree, the conclusion is robust to selective exposure. If they disagree, the substantive claim supported by one estimand may not hold for the other. When the identified set for a given estimand spans both signs, the distribution across quadrants and the implied selection parameters ($\rho_T - \rho_U$) indicate what patterns of selective

exposure would be required to sustain each conclusion. Theoretical knowledge of the setting helps assess which regions of the parameter space are plausible and which require difficult-to-justify selections. When panel or auxiliary data are available, point estimates can be located within the identified set to narrow the plausible range.

6. Report what the exercise reveals about the estimand you targeted. If the identified set confirms that the ATE and ATT share the same sign, report this as evidence that the conclusion is robust to selective exposure. If they diverge, report the identified set and discuss what it implies for the subpopulation the estimate is defined over. When there are strong theoretical or empirical reasons to believe exposure is random, the exercise serves a different purpose: it clarifies the epistemological stakes of that assumption by showing what conclusions would follow if it fails. In either case, the exercise makes explicit what the data are and are not consistent with, and allows the researcher and the reader to assess how much the substantive conclusion depends on the choice of estimand.

8 Conclusion

Most of what we know about the political consequences of violence from empirical causal research comes from counterfactuals in which everyone is exposed to a treatment that falls, and will always fall, on a select few. In this paper, I showed that the population-average effect and the effect among the exposed can yield opposite answers. The literature on criminal victimization has read positive population-average effects as evidence that violence produces a more engaged citizenry. Yet I find that those who become victims of crime were already more exposed than their neighbors before victimization, and that they react to it by retreating from everyday politics. Violence could well mobilize on average if everyone were equally exposed to it. But the victimized are often the vulnerable, and among them, violence is more likely than not to demobilize.

That targeting is selective and that the vulnerable are disproportionately exposed is well established in the theoretical literature on violence. What has not been established is the consequence

for empirical causal research: when these features coincide, the population over which an effect is defined determines its sign. Across criminal victimization in Latin America, civil war abduction in Uganda, and police contact in the United States, heterogeneous responses to violence likely coincide with selective exposure. A better understanding of who is exposed to violence and how they differ from those who are not should inform not only the design of empirical studies but also the interpretation of the estimates they produce.

Whether violence erodes or strengthens civic engagement among those who experience it determines whether affected communities can hold their governments accountable. The success of an intervention cannot be evaluated against a counterfactual that averages over people who would never have been exposed to it. This is true not only of violence but of any treatment that falls selectively and that we would not seek to universalize. For this class of treatments, the effect among the exposed is the quantity of first-order interest, for researchers and policymakers alike.

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